

Sanghyeok Moon 문상혁

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Research Interests: Efficient AI Systems, AI Agents, Language Models, Retrieval-Augmented Generation

EDUCATION

Mar. 2021 – **B.S. in Applied Mathematics, Kyung Hee University (Korea)**
Feb. 2027 (expected) – Double major: Computer Engineering
– GPA: 4.12/4.3 (4.35/4.5)

RESEARCH EXPERIENCE

Sep. 2026 – **Undergraduate Researcher, FlexML Lab**
Present Advisor: Prof. Insu Han, KAIST

Mar. 2026 – **Undergraduate Researcher, Efficient Multimodal Intelligence Lab**
Aug. 2026 Advisor: Prof. Sungha Choi, Kyung Hee University

- Memory-Agnostic Input Processing for LLM Agent Memory Systems

Nov. 2025 – **Undergraduate Researcher, Nonlinear Dynamics Lab**
Jun. 2026 Advisor: Prof. Junpyo Park, Kyung Hee University

- Echo State Networks (Reservoir Computing) for Time-Series Prediction

Jul. 2024 – **Undergraduate Researcher, BioMath Lab**
Aug. 2025 Advisor: Prof. Sunmi Lee, Kyung Hee University

- Agent-Based Modeling of Innovation Diffusion under Social and Political Interactions
- Large-Scale COVID-19 Contact-Tracing Networks for Network-Based Reproduction Number (R_t) Estimation
- SEIR-ABM Simulations across Multiple Network Topologies to Study the Effects of Outgroup Aversion on Vaccine Hesitancy

PUBLICATIONS

International Journals:

- [J2] Gira Lee*, KangHyun Yoon*, Sang Hun Lim*, **Sanghyeok Moon**, Sunmi Lee (2026)
“Innovation Diffusion under Political Identity Polarization: A Contextualized Agent-Based Approach”
SAGE Open (SSCI, Q1) Co-author
- [J1] Byul Nim Kim*, Junwoo Jo*, Chunyoung Oh, **Sanghyeok Moon**, Arsen Abdulali, Sunmi Lee (2025)
“A Novel Approach to Estimating R_t through Infection Networks: Understanding Regional Transmission Dynamics of COVID-19”
Frontiers in Public Health (SCIE, Q1) Co-author

RESEARCH PRESENTATIONS

- | | |
|------|---|
| 2025 | Impacts of Outgroup Aversion and Vaccine Hesitancy on Pandemic Spread
Poster Presentation, JSMB-ASMB (Kyoto, Japan) |
| 2024 | Time-varying Empirical Reproduction Number
Poster Presentation, KSIAM (Busan, Korea) |

PROJECTS

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| 2026 | Anything Note <ul style="list-style-type: none">Built a note-taking service that integrates multimodal lecture information into unified Memory Units, enabling natural-language search and evidence retrieval. Reasoning Diversity for Self-Distillation <ul style="list-style-type: none">Evaluated self-distillation objectives for improving reasoning diversity and mitigating strategy collapse in reinforcement learning for LLMs. |
| 2025 | RAG-based Legal QA Chatbot <ul style="list-style-type: none">Developed a legal QA chatbot using LangChain, Chroma, and GPT-4o mini, with retrieval-augmented responses grounded in case-law evidence. ABM-based Simulation of Fertility under Wealth Inequality <ul style="list-style-type: none">Built an agent-based simulation to analyze how wealth inequality and assortative mating affect marriage and fertility dynamics. Price Volatility Forecasting in Daejeon <ul style="list-style-type: none">Built a forecasting pipeline using 10 years of regional price data and applied ARIMA/SARIMA models to predict price volatility. |

AWARDS

- 1st Place**, KHU Nexus Innovation Challenge, NXT Cloud & AWS (2026)
- Popularity Award**, KHUDA Datathon (KHU'DATA), Kyung Hee University (2024)
- 1st Place**, Big Data & AI Data Analysis Competition, Kyung Hee University (2021)

SKILLS

Programming: Python, SQL

LLM: PyTorch, Transformers, LangChain, Chroma, vLLM

Modeling: Agent-Based Modeling, SEIR Models, ARIMA/SARIMA, Echo State Networks

Tools: NumPy, pandas, scikit-learn, Git, LaTeX

CERTIFICATIONS & LANGUAGE

- TOEIC: 960 / 990 OPIc: IM2
- SQL Developer (SQLD)